# 11 The Hardy–Weinberg Room

## From alleles to genotypes

| CORE PATH · BSC | EXTENSION · MSC | BEFORE THIS ROOM |
|---|---|---|
| **about 32 min** | **+ 28 min** | The Reproduction Room (10) made the gametes this room pairs up. |
| steps 1–4 | steps 5–7 | |

**OPEN THE ROOM**

`https://evolinus.github.io/zooe/teaching/evolutionary-laboratories/#hardyweinberg`

A single locus with two alleles, $A_1$ at frequency $p$ and $A_2$ at $q = 1 - p$. You build a parent population, pair its gametes at random, and compare the offspring with the $p^2 : 2pq : q^2$ that Hardy–Weinberg predicts. This guide asks what the theorem promises, and what it cannot.

### LEARNING OBJECTIVES

*After the core path you should be able to*

- Calculate expected genotype frequencies $p^2$, $2pq$ and $q^2$ from an allele frequency.
- Show that any starting mix of genotypes reaches Hardy–Weinberg proportions after one generation of random mating.
- Explain why a rare allele is found almost entirely in heterozygotes.
- Explain why a finite population misses the expected proportions by a little each generation.

*The extension adds*

- Predict the sampling error of offspring genotype frequencies from $\sqrt{(f(1-f)/N)}$.
- Test a population for departure from Hardy–Weinberg with a $\chi^2$ test.
- Connect the rare-allele result to why dominance ($h$) matters so much in the Selection Room.

### SUGGESTED ROUTES

- **35-minute practical (BSc):** steps 1–4 and questions 1–3. Students need a calculator.
- **Problem class:** steps 2 and 6 on the board, with the class building the population together.
- **MSc:** steps 5–7 and questions 4–6 (about 30 minutes).

**How to work through it.** Every step asks you to *predict* before you press anything, and to *explain* afterwards. Write the prediction down and commit to it: a wrong prediction you can account for teaches more than a right one you guessed. Runs are random, so where a step says so, run it more than once before you decide what happened.

## 1 Where the rare allele hides

8 min

**SET UP**

Close the panel *One generation of random mating* so the $p$ slider is active. $N = 50$.

**PREDICT FIRST**

At $p = 0.1$, calculate the three genotype frequencies. Of all the $A_1$ copies in the population, what fraction sit in heterozygotes?

**RUN**

Slide $p$ from 0.5 down to 0.1, watching the Hardy–Weinberg square and the *Genotype frequencies as p varies* chart.

**LOOK AT**

the three readouts $f(A_1A_1) = p^2$, $f(A_1A_2) = 2pq$ and $f(A_2A_2) = q^2$.

**EXPLAIN**

Why does a rare allele almost never appear in a homozygote? At what $p$ are heterozygotes most common, and how common can they be?

## 2 Any population, one generation

10 min

**SET UP**

Open the panel again. Drag the two handles to build a lop-sided population with few heterozygotes — for example 40% $A_1A_1$, 20% $A_1A_2$, 40% $A_2A_2$.

**PREDICT FIRST**

What is $p$ in this population? What genotype frequencies will the offspring have after one round of random mating?

**RUN**

Press **Mate randomly**.

**LOOK AT**

the three histograms: G0 observed, expected, and G1 after one random mating.

**EXPLAIN**

How could a population with so few heterozygotes produce so many in one generation? What does random mating "forget" about the parents?

## 3 Small populations miss the target

8 min

SET UP

Keep your lop-sided parents. $N$ = **10**, then $N$ = **200**.

PREDICT FIRST

How closely will the offspring match $p^2 : 2pq : q^2$ at each population size?

RUN

Press **Mate randomly** five times at each $N$, noting the heterozygote frequency each time.

EXPLAIN

The genotypes still follow the allele frequency within a generation. What is causing the misses, and which room is entirely about that?

## 4 Gamete by gamete

6 min

SET UP

$N$ = 10, any population.

PREDICT FIRST

When you build the offspring one gamete at a time, how is each offspring's genotype decided?

RUN

Press **Build G1 gamete by gamete** and watch the gamete wheel.

EXPLAIN

Why does drawing two independent gametes give exactly $p^2$, $2pq$ and $q^2$ as probabilities?

EXTENSION · MSC

## 5 How big should the misses be?

10 min

SET UP

$p$ = 0.5 in Hardy–Weinberg proportions; $N$ = 10, then 200.

PREDICT FIRST

The offspring heterozygote frequency is a sample of $N$ with expected value $f = 0.5$. Calculate $\sqrt{(f(1-f)/N)}$ at both sizes.

RUN

Press **Mate randomly** ten times at each $N$ and estimate the spread of the heterozygote frequency.

EXPLAIN

Does your measured spread match? Why do the realised allele frequencies drift too, by about $\sqrt{(pq/2N)}$?

## 6 Testing for Hardy–Weinberg

10 min

**SET UP**

Build a population with the handles, for example 30% $A_1A_1$, 10% $A_1A_2$, 60% $A_2A_2$, at $N = 100$.

**PREDICT FIRST**

Calculate $p$, the expected counts, and $\chi^2 = \Sigma(\text{observed} - \text{expected})^2/\text{expected}$ with one degree of freedom. Is the departure significant at 5% ($\chi^2 > 3.84$)?

**RUN**

Press **Mate randomly** and calculate $\chi^2$ for the offspring.

**EXPLAIN**

What biological processes could make a real population depart from Hardy–Weinberg? Which of them does one generation of random mating remove, and which would persist?

## 7 Recessive alleles and selection

8 min

**SET UP**

Close the panel. $p$ = **0.05**.

**PREDICT FIRST**

What fraction of individuals are $A_1A_1$? What fraction of $A_1$ copies are in heterozygotes?

**RUN**

Read the square and the readouts.

**EXPLAIN**

If $A_1$ carried a recessive advantage, how much of it could selection see? Connect this to step 6 of the Selection Room guide.

## DISCUSSION AND WORKSHEET QUESTIONS

1. BSc Cystic fibrosis affects about 1 in 2,500 births in some European populations. Assuming Hardy–Weinberg, estimate the frequency of carriers.
2. BSc "Dominant alleles become more common over time because they are dominant." Use the room to show why this is wrong.
3. BSc Which of the theorem's assumptions are you most confident fails in a real human population, and does it matter for the locus in question?
4. MSc Inbreeding increases homozygosity without changing allele frequencies. Explain how, and how $F$ modifies the Hardy–Weinberg proportions.
5. MSc The Drift and Selection rooms track only the allele frequency in diploid mode. Explain why this room licenses that shortcut, and when it would fail.
6. MSc Why can a population be in Hardy–Weinberg proportions at one locus and far from them at another?

## CHECKING YOUR REASONING

*Read this after you have written your own answers. These are what the model should show on most runs — not the only acceptable explanations.*

- **Step 1.** At $p = 0.1$: 0.01 $A_1A_1$, 0.18 $A_1A_2$, 0.81 $A_2A_2$. Nine $A_1$ copies in ten sit in heterozygotes (the fraction is $q$). Heterozygotes peak at 50%, at $p = 0.5$.
- **Step 2.** The parents have $p = 0.5$; the offspring come out near 25 : 50 : 25. Random mating forgets the parents' genotypes and keeps only their allele frequency.
- **Step 3.** At $N = 10$ the offspring scatter widely around the expectation; at $N = 200$ they land close. The misses are sampling — drift — which is the Drift Room's subject.
- **Step 4.** Each offspring gets two gametes drawn independently, each $A_1$ with probability $p$: both $A_1$ with $p^2$, one of each with $2pq$, both $A_2$ with $q^2$.
- **Step 5.** About 0.16 at $N = 10$ and 0.035 at $N = 200$.
- **Step 6.** $p = 0.35$; expected 12.25 : 45.5 : 42.25, so $\chi^2 \approx 25.7 + 27.7 + 7.5 \approx 61$ — a large departure. After one random mating it is typically below 3.84. Inbreeding, selection, assortative mating and population structure can all cause departures.
- **Step 7.** $p^2 = 0.0025$, so 1 individual in 400; 95% of $A_1$ copies are in heterozygotes, where a recessive advantage is invisible.